Emran-50
1st Tutorial
12.03.24

B.Sc. (Hons.) Part-III Tutorial Examination 2023
Course: Chem. 310, Marks: 20, Time: 1 hrs.

1. What do you mean by polar and nonpolar molecules? 2.0
2. Explain the terms distortion polarization and orientation polarization. 2.0
3. Derive an expression for Clausius-Mosotti equation. 4.5
4. Explain dipole-dipole, dipole-induced dipole and induced dipole-induced dipole interactions. 4.0
5. Describe the Ebert's and vapour-temperature method for the determination of dipole moment 3.0
6. The total molar polarization of CH_3Cl is 90.12 cc/mole at $1/T = 0.004$ and 50.0 cc/mole extrapolated to $1/T = 0.0$. Calculate the polarizability and dipole moment for CH_3Cl. 3.0
7. What do you mean by dipole moment? 1.5

Tutorial Examination-2023

Course: Chem-310; Time: 50min; Marks: 20

1. State and explain the laws of photochemistry. 2.5
2. What are the consequences of light absorption by molecules. 2.0
3. Differentiate between thermal and photochemical reactions. 2.0
4. Explain the terms fluorescence and phosphorescence using Jablonski diagram. 2.5
5. Explain the factors which are responsible for low and high quantum yield? 2.0
6. Deduce an expression for the kinetics of photochemical reaction between H_2 and Cl_2. 3.5
7. The dissociation energy of hydrogen is 103.5 kcal/mol. If H_2 is dissociated by illumination with [illegible] radiation of wavelength: 2537A°. What fraction of the radiant energy will be converted into K. E.?
8. Write down the applications of gas lasers, chemical lasers and dye lasers 2.5

B.Sc. (Hons.) Part-III Tutorial Examination 2023
Course: Chem-310; Time: 50min; Marks: 20

1. What is colloid? How does it differ from suspension?
2. What do you mean by sols? Distinguish between lyophilic and lyophobic sols.
3. Explain the Brownian movement of sol.
4. What do you mean by electrophoresis? Explain with diagram.
5. What is gel? Explain elastic and non-elastic gels and give suitable examples.
6. Write down the application of colloids.

312

Department of Chemistry
1st Tutorial Examination
Course Title: (Molecular Motion and Reaction Kinetics)
Full Mark: 10, Time: 30 min

1. (a) What is collision frequency? Derive an equation for collision frequency. 1+ 2
   (b) State Fick's first law of diffusion. 2
   (a) Define drift speed, ionic mobility and transport numbers. 3
   (b) Derive an expression for Einstein relation between the diffusion co-efficient and ionic mobility.

312

Second tutorial
Dept of chemistry
Total Mark: 10; Time: 25 Min

1. How Isotope effects in chemical reactions? Differentiate between primary and secondary isotope effects 1+2=
2. What is steady state approximation? 2+3=
   Derive the rate law for the decomposition of N_2O_5
3. Discuss Stopped-flow method to measure the changing concentrations of a fast reaction.

Department of Chemistry, Jahangirnagar University
1st Tutorial Examination-2023
Chem-315, Marks: 20, Duration: 1 hour

(i) What is meant by freezing point depression? Sketch a graph to show the depression of freezing point of a solvent due to addition of a solute. 3

(ii) Explain why freezing point depression of a solvent caused by 0.1 *m* NaCl is two times higher than that caused by 0.1 *m* sucrose. 1

2. If 0.44 g of a substance is dissolved in 22.2 g of benzene, then the freezing point of benzene is lowered by 0.567 ^{0}C. Calculate the molecular weight of that substance. Given that K_f is 5.12^{0}C mol^{-1}. 4

3. (i) Define adsorption isotherm. Write the mathematical expressions of Langmuir and Freundlich adsorption isotherms. 3

(ii) How could you show the validity of Langmuir adsorption isotherm? 1

4 Define partial molal volume. Draw a specific volume versus mole fractions of ethanol for water-ethanol mixed solvents. Using this plot, how could you determine the partial molal volume of ethanol? 4

5 What is meant by specific rotation? Discuss the principle of determining the concentration of sucrose solution from the measurement of specific rotation. 1.5

320

In valence bond theory (VBT) the wave function for a system containing two non interacting hydrogen atoms is given as $\Psi A = \Psi A(1)\, \Psi B(2)$, where ΨA and ΨB designate the wave functions of atoms A & B and the numbers 1 and 2 designate the two electrons. 2+2+3=7

(i) Write the wave function when the electrons are exchanged to allow for covalency.

(ii) Give the wave function when both the electrons remain on atom A or B giving ionic structure.

(iii) Explain, by drawing the potential curves, how do the energies change in (i) and (ii).

(b) Define hybridization. Why is the use of hybrid orbitals are preferable over pure atomic orbitals in forming covalent bond? 3

2. (a) What do you mean by Molecular Orbital? How does it differ from Atomic orbital? 2

(b) Explain with suitable example the formation of MO from AO through LCAO.

(c) Draw the bonding and antibonding orbital resulted from 2p orbitals with

(i) σ interaction (ii) π interaction.

(d) Draw the MO diagram of CO molecule. Hence explain how it can act as π acid ligand. 1+

320

(1) Explain coordination number and coordination sphere with suitable example.

(2) Conductivity of $CoCl_3.6NH_3$ is 432 $ohm^{-1}cm^2mol^{-1}$ while that of $CoCl_3.5NH_3$ is 261 o

- Explain. How many mole of AgCl will be formed from each complex?

(3) Calculate the effective atomic number of the central metal atom of the following comp hence comment on their stability. (i) $[Mn(CO)_5]$, (ii) $[Mn(CO)_5]^-$,

(4) Draw the structure of:

(i) Linkage isomers of $[Co(NH_3)_5NO_2]^{2+}$ (ii) ionization isomers of $[Co(en)_2(NCS)$

(5) Chelate complexes are generally more stable – explain.

(6) Explain the formation of the following on the basis of VBT:

(i) paramagnetic $[FeF_6]^{3-}$ (ii) diamagnetic $[Ni(CN)_4]^{2-}$ (iii) tetrahedral

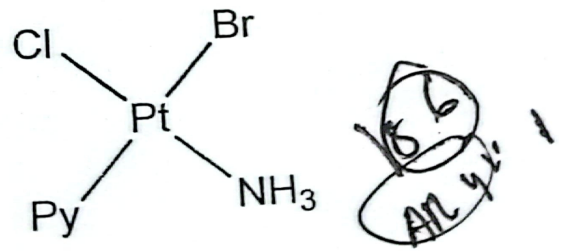
(1) Explain the splitting of d orbitals of metals in coordination compounds. Draw the d-orbital splitting pattern for octahedral, tetrahedral and square planar complexes 2+3 =

(2) Determine the number of unpaired electrons and calculate the CFSE (ignore the pairing energy) of the following: $[Cr(H_2O)_6]Cl_2$, $[Ni(NH_3)_6]^{2+}$, and $K_3[Co(CN)_6]$. 3x2=

(3) Mode of splitting of d-orbitals in an octahedral field is just reverse of that in tetrahedral field – explain. 3

(4) What is trans-effect? 2

(5) Using K_2PtCl_4 and other appropriate reagents, show by reaction scheme how will you prepare the compound below: (Given that trans effect follows the order Br->Cl-.py>NH_3) 4



Department of Chemistry
1st Tutorial Exam., Course No 322
Time : 40 min., Marks -20

1. a) Distinguish among conductors, semiconductors and insulators on the basis of band theory of solids. 5

(b) As the temperature increases the electrical resistance of a metal increases – explain. 2

(c) Briefly describe the intrinsic and extrinsic semiconductors? 3

2. (a) Why is B_2H_6 an electron deficient compound? 2

(b) Draw the structures of B_5H_9 and $B_{10}H_{14}$ and discuss the unusual bonding in diborane. 2+3 =5

(c) Why is borazine called the inorganic analogous of benzene? 3

Department of Chemistry
Course -322
2nd Tutorial Examination
Time- 40 min, Marks- 10

1. Compare the solvent properties of H_2O and liquid NH_3 3

2. Describe the different types of Chemical Reactions taking place in Liq. NH_3 5

3. The solution of alkali metal in Liq. NH_3 is colored- explain. 2

Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 40 min.

1. (a) Define *symmetry operation* and *plane of symmetry*. What is the difference between *vertical plane* and *horizontal plane*?

1 × 2 + 1 = 3

(b) Using an appropriate flow chart determine the point group of $SOCl_2$, SF_4 and C_6F_6. **1.5 × 3 = 4.5**

(c) What is the condition that needs to be fulfilled if the two symmetry operations belong to the same class.

1

(d) Taking the C_2 axis as the z axis, and σ_h to be the xy plane, to what representations do the P_x, P_y, and P_z orbitals belong in C_{2h} symmetry (the Character Table of C_{2h} point group is given below, in part, for your convenience).

1.5

C_{2h}	E	C_2	i	σ_h
A_g	1	1	1	1
B_g	1	-1	1	-1
A_u	1	1	-1	-1
B_u	1	-1	-1	1

Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 45 min.

1. Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the 2*s* and 2*p* orbitals of carbon and the *1s* orbital of each hydrogen atom. The Character Table of T_d point group is given below for your convenience: **6**

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	(xz, yz, xy)

2. Find the number of C–O stretching vibrations which are active in the infrared and Raman spectra of *cis*-$[M(CO)_4L_2]$. The Character Table of C_{2v} point group is given below for your convenience: **4**

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma'_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

1st Tutorial Exam, Course No.**Chem 330**.Date; 02.05.24

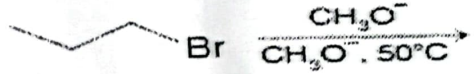
1. What products would you get when trans-2-butene and cis-2-butene reacts with bromine? 4
2. What is a stereospecific and regioselective reaction with example? 2
3. Why Trans-1- phenylpropene give only 70% ANTI-addition product when react with bromine? 4
4. Define S_N1, S_N2 and S_Ni reactions with mechanism. 6
5. The rate of solvolysis of the tertiary halide, Me_3CBr is found to be $3x10^4$ times faster in 50% aqueous Ethanol than in ethanol – Explain 4

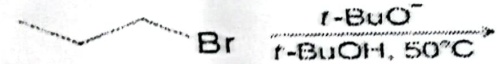
and Starch. 3.0

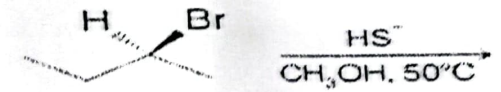
, 2825, 1755, 1470, 1370, 1200 cm .

2nd Tutorial Exam, Course No.**Chem 330**.Date; 12.09.24

1. Write down the mechanism for E2 and E1cB elimination reactions. 4
2. Explain why $MeCH_2CMe_2Cl$ is found to yield only 34% of alkene whereas Me_2CHCMe_2Cl is found to Yield 62%of alkene. 3
3. In most simple acyclic cases, ANTI elimination is found to be very much preferred-Justify the statement. 4
4. Discuss Hofmann and Saytzev elimination rule. 4
5. Give the product (or products) that you would expect to be formed in each of the following reactions; 5

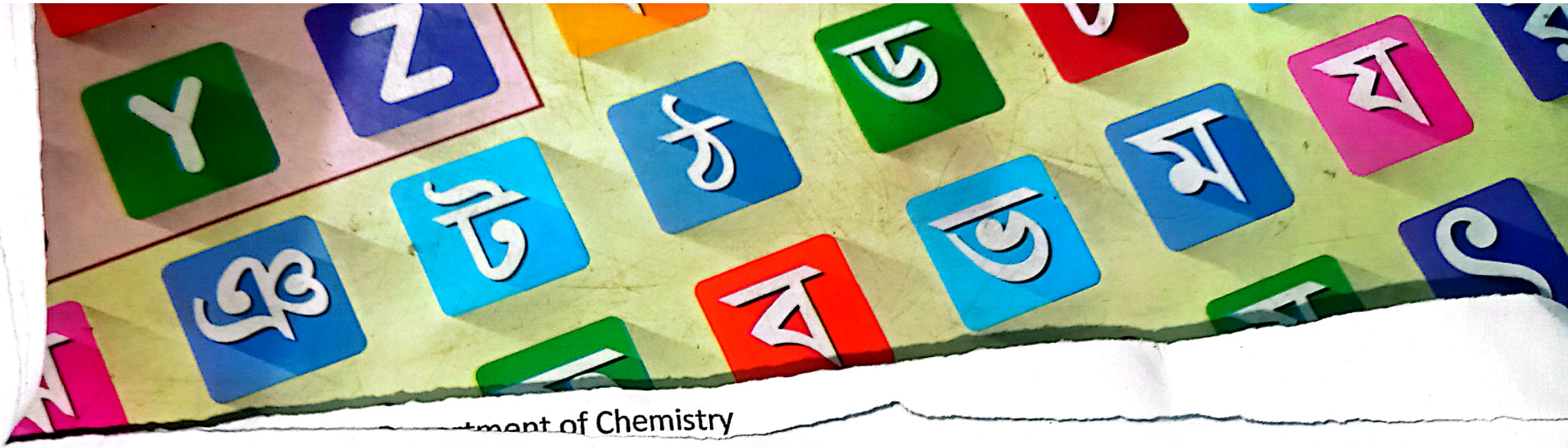
(a) 

(b) 

(c) 

(d)

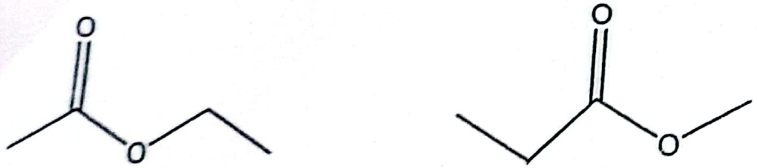
(e)



tment of Chemistry

3rd Tutorial Exam, Course No. **Chem 330**. Date; 07.11.24

1. Write down the criteria for IR activity. Why C=O group of ester absorbs at higher frequency than ketone in IR spectrum?
2. Why nmr spectrum of benzene is observed at a lower field whereas that of acetylene is observed at a higher field strength?
3. Explain why auxocrome are producing a bathochromic shift?
4. An ester having a molecular formula $C_4H_8O_2$ can have the following two structures; 4



A

B

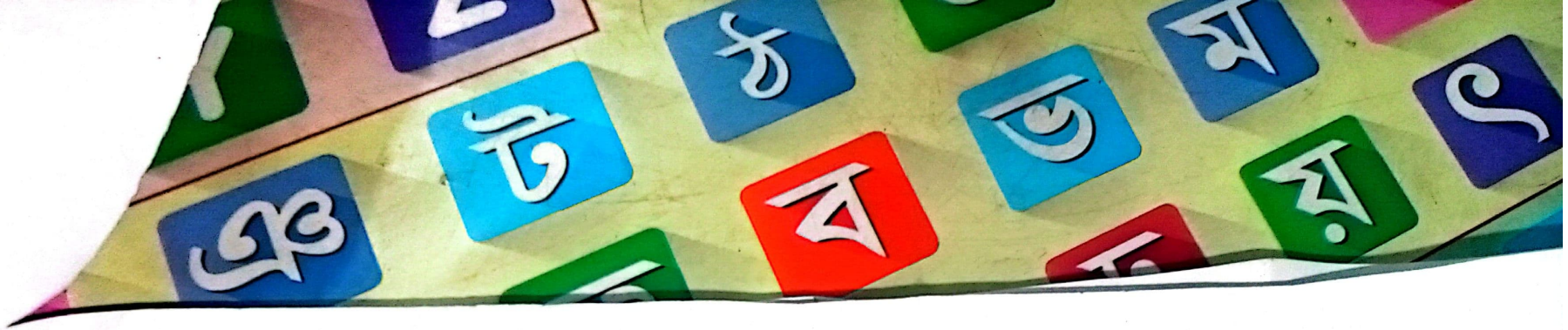
Which structure fits the following data: δ_H=1.2(3H,t), 2.05(3H,s), 4.15(2H,q)? Give reasons

5. Define metastable ion peak and Maclafertty rearrangement. Show the fragmentation pattern for 2-pentanone.

molecular

Department of Chemistry
3rd year tutorial exam-1, 2023
Course no: 332, Chemistry of natural products
Unit=1, Marks= 20, Time 40 mints

1. Classify amino acids according to the properties of their side chain. 3
2. Explain why the acidity and basicity constants of amino acid are ridiculously low? 2
3. Write down the Gabriel phthalimide and Streaker synthesis of amino acids. 3+
4. Define peptide. Synthesize the peptide Tyr-Gly-Ala. 2+
5. What is β pleated sheet of protein?



Department of Chemistry

3rd year Tutorial examination 2023

Course no: 332, Course Name: Chemistry of natural products

a. Explain the Herzig-Meyer method for determining N-CH_3 groups in alkaloids. 3

b. Why is L (+)-Ephedrine more basic than D (-) Ephedrine? 3

c. The 'S' Sulphur in Penicillin is not in thiol form but it is in a five-membered ring. Explain with necessary reactions. 4

d. Outline the synthesis of Chloramphenicol. 3

e. How can you identify 3°amine in cyclic form in alkaloids? 4

f. Write down different mechanisms of action, an antibiotic can follow against bacteria. 3

... and Starch. 3.0

... 2930, 2825, 1735, 1470, 1370, 1200 cm^{-1}.

^{1}H-NMR; δppm0.9 (3H,t), 1.6 (4H, m), 2.05 (3H, s), 4.15 (2H, t)

... the transition for a two-electron atom whose spin is paired. 5.0

Department of Chemistry
Jahangirnagar University
Course No. Chem. 332F
Title: Chemistry of Natural Products; Marks: 20
Tutorial 1, Time: 45 min., Date: 09-09-2024

Emran
chem-50

1.	(a) Define epimer and mutarotation with suitable examples.	3
	(b) Distinguish between reducing and non-reducing sugars? Show that (+)-Maltose is a reducing sugar whereas (+)-Sucrose is a non-reducing sugar. Explain it with proper reactions.	2+3
	(c) What is osazone? Show that the D-(+)-Glucose and D-Mannose give the same osazone. Explain it with proper reactions.	2+2
	(d) Give structures and names of the products of the reactions of D-(+)-glucose with: (i) Nitric acid (II) HIO_4 (iii) CH_3OH, HCl; then $(CH_3)_2SO_4$, NaOH; then dil. HCl	5.0
	(e) Discuss about the conformation of (+)-Maltose, (+)-Sucrose and Starch.	3.0

IR; λmax 2950, 2825, 1735, 1470, 1370, 1200 cm^{-1}.

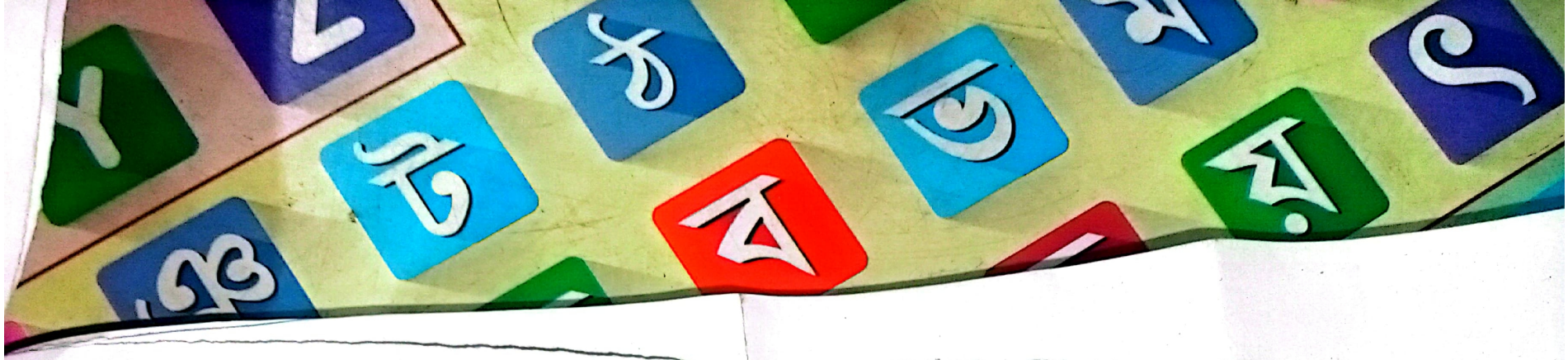
^{1}H-NMR; δppm0.9 (3H,t), 1.6 (4H, m), 2.05 (3H, s), 4.15 (2H, t)

...electron atom whose spin is paired.

3-1 = 2

2+3/2 = 2-

5.0



Tutorial Examination, Course no; 335, Marks; 20, Time; 40 min

1. Why is the sodium extract alkaline in nature? 2
2. How will you detect the elements "N"and "S" by sodium fusion method? 4
3. Mention the solubility class for each in the following compounds: 3

   a) Acetophenone b) p-ChloroBenzoic acid c) D- Glucose d) Amino phenol

   e) Naphthalene f) 4- methylaniline
4. How can you differentiate between the following pairs by chemically?Give 5 necessary equations.

   a) Aniline and methylamine b) Pentanal and Pentanone
5. An unknown compound has the following IR and ^{1}H-NMR data, predict the **name And structure** of this unknown compound and also **explain each of the spectral value.** 6

   IR; λmax 2950, 2825, 1735, 1470, 1370, 1200 cm^{-1}.

   ^{1}H-NMR; δppm0.9 (3H,t), 1.6 (4H, m), 2.05 (3H, s), 4.15 (2H, t)

Explain the transition for a two-electron atom whose spin is paired.

Department of Chemistry
1st Tutorial Examination-2024
Course no. Chem-340
Date: 19/05/2024

Time: 50 min

Marks:

1. What is meant by molecular spectroscopy? Write the name, region, and molecular spectroscopy of different electromagnetic radiation.
2. Using a suitable block diagram write the basic elements of absorption and emission spectroscopy.
3. Explain the factors that are responsible for the intensity of the spectral line.
4. Write the expression for the transition of a rigid and non-rigid diatomic linear molecule. Which rotational energy levels are much exaggerated and why?
5. Does isotope substitution cause a molecule to microwave active? Write the effect of isotope substitution in a molecule with a suitable diagram.

...between NH_3 and BF_3 using 4.0

...angular momentum of 3/2, 1 and orbital

...and explain the transition for a two-electron atom whose spin is paired.

Department of Chemistry
2nd Tutorial Examination-2024
Course no. Chem-340
Date: 27/10/2024

Time: 50 min Marks: 20

1. When does an anharmonic oscillator behave like a harmonic oscillator? Calculate the absorption energy of the fundamental band, first overtone, second overtone, and hot band for an anharmonic oscillator. 4.0
2. How many modes of vibration are possible for H_2O molecules? Explain the reason which modes are IR active or inactive. 4.0
3. A mixture in which normal and isotopic CO_2 are present. How do you distinguish them using IR spectroscopy? 4.0
4. In CO_2 molecules, all the fundamental modes of vibration change the polarizability ellipsoids; however, not all are Raman active. Justify the statement. 4.0
What is meant by the law of mutual exclusion? How do you distinguish between NH_3 and BF_3 using Raman and infrared spectroscopy? 4.0

... angular momentum of 3/2, 1 and orbital ...
... and explain the transition for a two-electron atom whose spin is paired. 5.0

Department of Chemistry
3rd Tutorial Examination-2024
Course no. Chem-340
Date: 05/12/2024

Time: 50 min

Marks: 20

1. What are the factors that govern the energy of the atomic orbitals? Calculate the energy of the orbitals when an electron remains in the 1s and 2p state. 5.0
2. Explain the fine structure of the hydrogen atom spectrum. Which transition gives rise to the compound doublet spectrum? 5.0
3. What is the value of L, S, and J from the following term symbol?
   i) $^4S_{3/2}$ ii) $^2D_{5/2}$ iii) $^4P_{3/2}$
   Write the term symbol for the electrons with a total spin angular momentum of 3/2, 1 and orbital angular momentum of 2, 3 respectively. 5
4. Draw and explain the transition for a two-electron atom whose spin is paired. 5

3rd year B.Sc. (Hons.) Tutorial Examination 2024
Department of Chemistry
Jahangirnagar University, Savar, Dhaka
Course No. : CHEM 350
Course Title : Fundamental of Pharmaceutical Chemistry
Time : 30 mints, Full Marks : 10

1 **(a)** What is the most popular oral dosage form and why?

(b) Briefly discuss about tablet granulation process.

(c) What is the Coating of tablets? What are the coating defects?

(d) How many classes of capsule? Discuss about it.

(e) Calculate the LOD and Moisture content of a drug when mass of drug 100mg and mass dry drug after drying 75mg.

3rd year B.Sc. (Hons.) Tutorial Examination 2024
Department of Chemistry
Jahangirnagar University, Savar, Dhaka
Course No. : CHEM 350
Course Title : Fundamental of Pharmaceutical Chemistry
Time : 30 mints, Full Marks : 10

1 **(a)** What are the pharmaceutical raw materials? Classify and define. 2

(b) Write the examples of the following classes of API. 2
i) Antibiotic ii) Cardiovascular iii) Gastrointestinal iv) Anti diabetic

(c) Classify receptor according to their mechanism of action. 2

(d) What are the routes of drug administration? 2

(e) Define Reference standard and working standard in pharmaceutical industry. 2

3rd year B.Sc. (Hons.) Tutorial Examination 2022
Department of Chemistry
Jahangirnagar University, Savar, Dhaka
Course No. : CHEM 350
Course Title: Pharmaceutical Chemistry
Time: 30 mints, Full Marks: 10

1	**(a)**	Define QA and QC.	2
	(b)	What are GMP and cGMP? Write down important principles of GMP.	2
	(c)	What is the principle of the Karl Fischer titration? How can you calculate % of water by using Karl Fischer reagent.	2
	(d)	An assay experiment, 30.30 mL of N/10 I_2 solution required to titrate 300 mg of ascorbic acid. The normality factor of iodine solution is 1.106. Calculate the percentage content of L-ascorbic acid in supplied sample.	2
	(e)	What do you mean by HPLC? Write down the application of HPLC in pharmaceutical industry.	2